Problem 26.9

What is the present value of a 30 year immediate annuity at 10% interest where the first payment is \$100 and each payment thereafter is increased by 5%?

Solution

The payments form a geometrically increasing annuity-immediate:

$$100,\ 100(1.05),\ 100(1.05)^2, \ldots, 100(1.05)^{29}.$$

The first payment is

$$C_1 = 100,$$

the annual effective interest rate is

$$i = 0.10,$$

and the growth rate is

$$g = 0.05.$$

For a geometrically increasing annuity-immediate, the present value is

$$PV = C_1 \left(\frac{1 - \left(\frac{1+g}{1+i}\right)^n}{i - g} \right).$$

Substitute $C_1 = 100$, $g = 0.05$, $i = 0.10$, and $n = 30$:

$$PV = 100 \left(\frac{1 - \left(\frac{1.05}{1.10}\right)^{30}}{0.10 - 0.05} \right).$$

Thus,

$$PV = 1504.631417.$$

Therefore, the present value is

$$\boxed{1504.63}.$$